

SW2

EventLens: Event Risk Research for Indices and Equities

by

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Abstract

This proposal presents EventLens, a research system that tests whether prediction-market activity can improve event-driven risk analysis for major US indices and selected index-tracking exchange-traded funds (ETFs). Polymarket, a prediction market, provides market-implied event probabilities as prices, while on-chain settlement records link activity to wallet addresses. EventLens combines these sources to estimate experimental wallet-informed probabilities.

The study first compares market and wallet-informed probabilities on historical events. It then tests whether prediction-market features improve volatility forecasts against a recent-return baseline. Subject to data availability, an extension will test whether reviewed related-event features and agent-assisted event selection add incremental out-of-sample value.

EventLens presents the analysis through a dashboard and a controlled artificial-intelligence assistant. Users can compare probability estimates, inspect supporting wallet evidence, explore portfolio scenarios and receive explanations linked to versioned data and model outputs.

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1 Introduction

1.1 Background and Motivation

Financial markets are repeatedly repriced around discrete events such as central-bank decisions, elections, and geopolitical developments. Investors can observe news, asset prices, and volatility indicators, but these sources do not directly show a continuously updated probability of the underlying event. This project aims to develop a machine-learning-assisted quantitative analysis framework, EventLens, that extracts information from prediction markets and evaluates its value in forecasting financial-market volatility.

Prediction markets. Prediction markets such as Polymarket aggregate the beliefs of market participants into continuously updated probabilities. Market participants can trade YES and NO shares on the occurrence of events. A YES share pays one dollar if the specified outcome occurs and zero otherwise. A price of 60 cents can therefore be interpreted as an approximate market-implied probability of 60% [4, 5].

Wallets and blockchain records. Web3 prediction markets such as Polymarket record trade settlements on the blockchain. Their blockchain-based infrastructure provides transparent transaction histories that enable the behavior and historical performance of individual participants (identified by their wallet addresses) to be analyzed. This creates an opportunity to investigate whether information embedded in prediction-market activity, particularly from informed traders or "smart wallets", could provide useful signals for traditional financial markets.

1.2 Objectives

1. **Develop a smart-wallet identification model** using historical on-chain trading behavior to identify addresses whose decisions may add information beyond the market price.
2. **Construct wallet-informed event probabilities** using wallet evidence and compare these experimental estimates with market-implied probabilities.
3. **Develop an event-relevance framework** using statistical methods and a controlled LLM research agent to identify both relationships among prediction-market events and their relevance to selected indices and ETFs.
4. **Evaluate whether Polymarket data provides incremental and time-stable predictive value** through walk-forward backtesting, ablation studies and comparison against benchmark volatility forecasts, and present the results through an interactive AI-assisted dashboard.

The primary target user is a research-oriented retail investor or finance professional who understands basic portfolio concepts but lacks the infrastructure to query blockchains, evaluate prediction-market participants, and construct event-conditioned risk models. EventLens is intended as a research and decision-support tool for hypothetical portfolios rather than personalized financial advice.

Scope

This project will be divided into two phases. The first phase will focus on the minimum viable product (MVP), to validate the core concept, while the second phase will explore extensions and additional features. Table 1 summarizes the project's minimum viable scope and potential extensions.

Table 1: Minimum viable scope and extended scopes.

Dimension	MVP commitment	Extensions
Venue	Polymarket on Polygon	Cross-platform comparison, for example, with Kalshi
Event coverage	Federal Reserve rate decisions	Other macroeconomic and geopolitical events
Assets	Selected index-tracking ETFs	Individual equities and user-defined multi-asset portfolios
Horizons	1, 5 and 20 trading days	Additional term-structure horizons
Wallet model	Address-level, past-only scoring	Validated entity clustering
Event relationships	Target-event features and reviewed event–asset mappings	Experimental discovery and aggregation of related Polymarket events

1.3 Literature Survey

Prediction markets and informative participants

Research on prediction markets suggests that prices may be disproportionately influenced by a relatively small group of informed or disciplined participants, suggesting that identifying these participants could provide valuable insights. Oliven and Rietz [9], for example, find that active market makers in the Iowa Electronic Markets behave more consistently with rational pricing conditions than typical price takers. This supports the “marginal trader” hypothesis: market accuracy may depend less on the average participant than on the traders willing and able to act against mispricing. Evidence from forecast aggregation is more mixed. Chen et al. [3] find that weighting forecasters according to past performance does not significantly outperform simpler pooling methods in their sports-prediction sample. By contrast, Atanasov et al. [1] show that performance weighting, temporal decay and recalibration can produce forecasts that outperform standard prediction-market prices.

Recent Polymarket research provides more direct evidence of participant heterogeneity. Akey et al. [2] find that approximately 69% of users lose money and that the top 0.1% capture more than half of aggregate gains. Gómez-Cram et al. [8] classify approximately 3% of accounts as skilled. These accounts do not hold insider information but exhibit out-of-sample persistence and respond more rapidly to public information.

Prediction-market price as a proxy of event probability

Binary prediction-market contracts pay one unit if a specified outcome occurs and zero otherwise. Their prices are therefore commonly interpreted as event probabilities. Wolfers and Zitzewitz [4] argue that this interpretation can be a useful approximation under risk neutrality, limited risk aversion and sufficiently liquid trading. Nevertheless, the observed price remains an equilibrium outcome determined by traders’ beliefs, wealth, preferences and the market mechanism, not automatically the physical probability that the event will occur.

Manski [14] demonstrates this limitation formally. With heterogeneous risk-neutral price-taking traders, the equilibrium price represents a budget-weighted quantile of the belief distribution rather than its mean. Consequently, the price may bound the average belief without identifying it exactly. Gao et al. [16] extend the analysis to heterogeneous risk-averse traders and utility-based market makers. Depending on the utility specification, the limiting price may resemble a geometric or power mean of beliefs and also depend on risk aversion, initial wealth and trading behavior.

2 Methodology

2.1 Design

System overview

EventLens is a Polymarket-informed research system for forecasting volatility and examining event-conditioned portfolio risk. It acquires prediction-market, indexed settlement and conventional financial data, preserves the evidence with its original timing, and converts it into versioned research features. Analytical services identify historically informative wallets, estimate market and wallet-informed event probabilities, map events to supported assets, and compare baseline and augmented volatility forecasts. The same validated calculations drive portfolio scenarios, the investor dashboard and a controlled research assistant. Walk-forward evaluation determines which model versions may be used. User selections and reviewed relevance feedback shape an analysis, but cannot change wallet eligibility or fitted results.

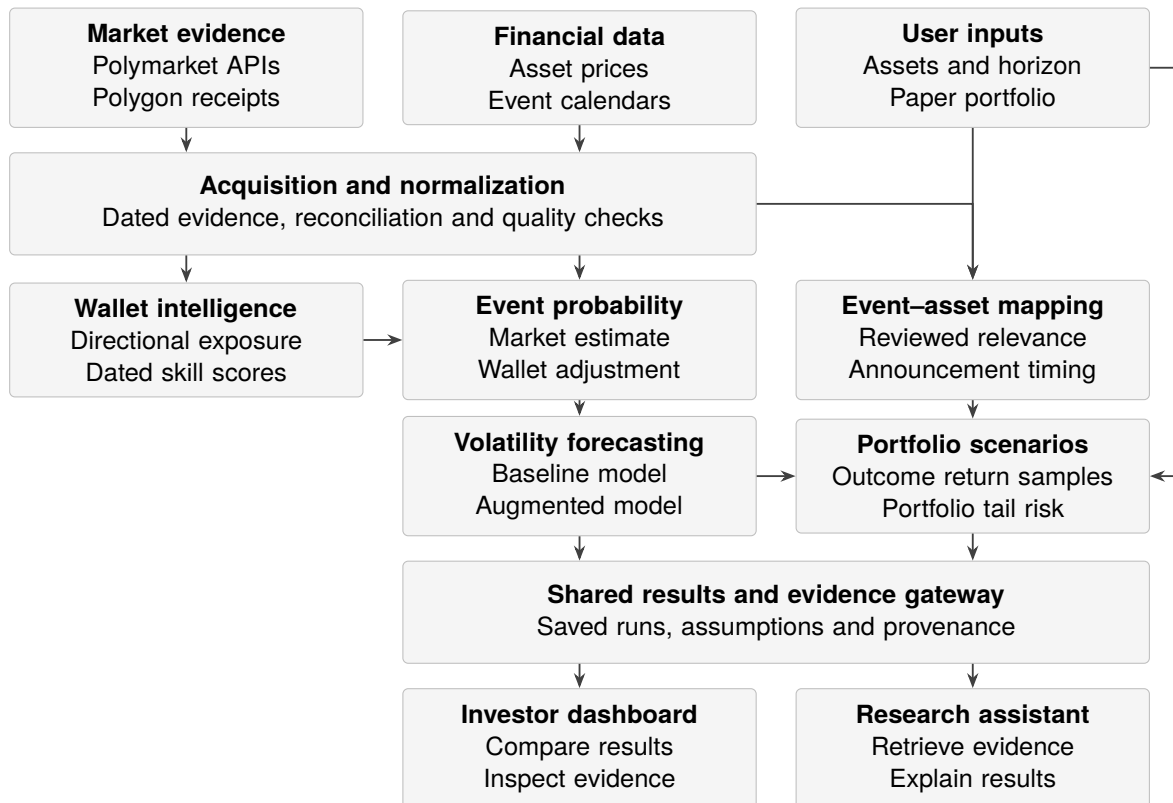


Figure 1: Logical system architecture. Arrows carry evidence or calculation results.

Components and responsibilities

Table 2: Logical system components and their responsibilities.

Component	Principal inputs and responsibility	Principal outputs
Acquisition and normalization	Market APIs, indexed settlement records, Polygon receipts and financial feeds. Normalize records and align as-of cut-offs.	Immutable raw and time-series evidence with quality and provenance flags.
Wallet intelligence	Past wallet decisions, reconstructed exposure and resolved outcomes. Estimate historical informativeness.	Dated eligibility, skill, concentration and confidence measures.
Probability and relevance	Market state, wallet signals, event rules, asset coverage and reviewed mappings.	Raw and wallet-informed probabilities plus event-asset mappings.
Forecasting and scenarios	Versioned event features, asset returns, baseline volatility and paper portfolios.	Volatility forecasts, uncertainty, scenario losses, VaR and expected shortfall.
Delivery and explanation	Calculation results, evidence links and user context.	Dashboard views and supported assistant answers.

Data Acquisition and normalization

The data platform combines prediction-market information, public blockchain settlement records and conventional financial data. These sources serve different purposes.

Table 3: Planned sources, with access decisions made during the feasibility audit.

Source	Data to acquire	Use and limitation
Polymarket metadata API (Gamma)	Event definitions, rules, identifiers, outcome prices and available price history.	Define the forecast target and archive changes to mutable metadata [16].
Polymarket CLOB and activity Data API	Bid/ask snapshots, liquidity and public wallet activity.	Measure market quality and candidate wallet flow. Historical book depth may need prospective collection.
Goldsky Polymarket datasets	Indexed fills, matched trades and relevant position records.	Targeted historical reconstruction, subject to version coverage and access cost [17].
Polygon RPC	Transaction receipts, contract logs and block references.	Verify a stratified sample of settlements. Older state may require archival access.
Alpha Vantage or an approved equivalent	Daily equity/ETF prices, splits and dividends, with intraday data if available.	Proposed primary price-provider adapter. Adjusted daily data are a premium function. Coverage and academic-use terms must be checked [18].
Federal Reserve	Meeting dates, statements and announced rate decisions.	Establish announcement timing and validate event labels separately from platform resolution [19].

Logical data model

Figure 2 groups records by domain. Solid arrows show one-to-many relationships. Dashed arrows show analysis dependencies. Section 2.2 specifies the core physical keys and fields.

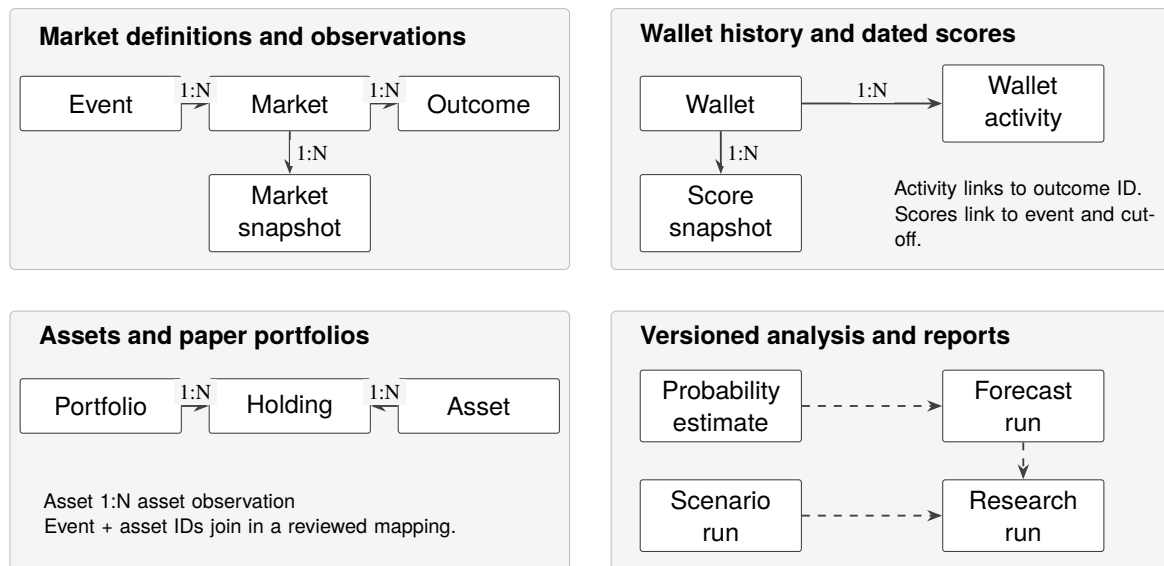


Figure 2: Principal records grouped by domain. Cross-domain references use event, outcome, asset and run identifiers.

Select and track informative wallets

An informative "smart wallet" is an observed account whose previous decisions contain information beyond the market price. The first score will examine whether the wallet repeatedly chose the eventual outcome at informative prices, whether prices subsequently moved in its direction, and how broad its prior history was. Note that a wallet may be informative for one event but not another, and that a wallet's informativeness may change over time. The score will therefore be dated and event-specific, with a separate eligibility test for each forecast cut-off.

Repeated fills will be consolidated into wallet–event decisions. Balanced YES and NO holdings will be distinguished from directional exposure. Scores based on few independent events will be pulled towards the population average, a technique called shrinkage. A provisional eligibility rule is ten earlier resolved events across at least three announcement dates, with five- and twenty-event thresholds checked in sensitivity analysis. If history cannot support this rule, the wallet signal will be marked unavailable rather than assigned an unsupported confidence score.

Wallet eligibility and scores will use only records available and outcomes resolved by each forecast cut-off. New activity updates exposure, while newly resolved events update only later scores. Frozen snapshots prevent today's successful wallets from being retrospectively selected for earlier tests.

Estimate market and wallet-informed probabilities

The raw event probability will begin with the YES bid/ask midpoint, recording spread, depth and timestamp. When both outcome books are usable, their midpoints can be normalized for consistency. A missing or one-sided book will be flagged rather than silently replaced by an unrelated latest trade.

A regularized logistic model will calibrate the raw estimate using earlier resolved events and add a small number of wallet signals: overall directional flow, the flow of historically informative wallets and market-quality controls. Coefficients will be fitted on training data rather than chosen to produce a desired wallet-informed probability. Beta calibration is a possible robustness check if justified by the sample [14]. The raw and wallet-informed probabilities will remain separately visible so the wallet contribution can be

evaluated directly.

Link events to assets

The event research agent may suggest that a rate decision matters to SPY or MSFT, but numerical impacts will come from matched historical event windows and conventional market controls. The mapping must account for surprise relative to the expectation already embedded in prices. For the MVP, we will explore using an AI agent to suggest candidate mappings, then review and approve them manually by the user.

Related-event extension. Different Polymarket contracts may contain overlapping information because they refer to the same entity, topic, time period or underlying outcome. Subject to data availability, EventLens will test whether features from reviewed related events provide incremental predictive value beyond the selected target event. This related-event analysis is an optional extension. Candidate relationships may be proposed from timestamped contract metadata by the research agent, but will be retained only after deterministic or manual review. The study will interpret these relationships as predictive associations and will not claim causality without a separate causal identification design.

Agent ablation. The contribution of the research agent will be evaluated by comparing three configurations: fixed rule-based event selection, agent-proposed event selection and a hybrid configuration in which agent proposals undergo deterministic or manual review. All configurations will use the same point-in-time data, forecast dates and downstream statistical model. The LLM will propose and explain relationships; numerical forecasts will continue to come from reproducible calculation tools.

Forecast volatility

After estimating event probabilities, EventLens will test whether prediction-market information improves a simple forecast of asset-price variation. The comparison uses one regression baseline and an augmented version of the same model.

The forecast target over H trading days will be the sum of squared daily log returns:

$$RV_{t,H} = \sum_{h=1}^H r_{t+h}^2. \quad (1)$$

This daily-data proxy for realized variance measures the size of price movements without treating gains and losses as cancellations, but is noisy, especially for one day. Comparisons use the same horizon and units. Displayed volatility is the square root of the corresponding variance estimate.

Baseline. A linear regression with an intercept and the average squared daily return over the preceding 20 trading days provides a simple forecast based only on recent asset-price variation. A separate ordinary-least-squares fit will be made for each supported asset and horizon.

Augmented model. The same regression retains that input and adds the market-implied probability p_{raw} and wallet-informed probability difference $p_{\text{adj}} - p_{\text{raw}}$. Both versions use identical forecast dates, targets and training windows restricted to dates with usable event evidence. The augmented model therefore tests the combined contribution of prediction-market information, while separate probability experiments test whether wallet histories themselves add information.

Forecasts use the same small positive floor, fixed before testing. Event features for training and evaluation are generated only from data and probability-model fits available before each forecast date. Mean squared error (MSE) and quasi-likelihood loss (QLIKE) compare later, unseen observations [15]. The interface displays this fixed comparison rather than allowing users to change the baseline or feature set during

evaluation.

Explore portfolio scenarios and tail risk

For a selected portfolio, the engine combines conditional return distributions for the YES and NO outcomes. Let p be the event probability, and let μ and σ^2 be the portfolio mean and variance over the same horizon:

$$\mu_{\text{mix}} = p\mu_{\text{YES}} + (1 - p)\mu_{\text{NO}}, \quad (2)$$

$$\sigma_{\text{mix}}^2 = p\sigma_{\text{YES}}^2 + (1 - p)\sigma_{\text{NO}}^2 + p(1 - p)(\mu_{\text{YES}} - \mu_{\text{NO}})^2. \quad (3)$$

The last term captures uncertainty about which outcome occurs. Consequently, a higher adverse-event probability can worsen expected loss without necessarily increasing variance. For multiple holdings, the engine preserves cross-asset dependence when sampling returns, then applies the user’s weights to each joint scenario.

Outputs include scenario returns, horizon volatility, a chosen loss-threshold probability, 95% VaR and 95% expected shortfall. VaR is a threshold, not the maximum loss. Expected shortfall is the average of the worst 5% of the weighted loss distribution, including fractional threshold mass where needed. Empirical or heavy-tailed scenarios will be checked for sensitivity to probability and asset-response assumptions.

Controlled assistant roles

The research assistant has three roles within one controlled workflow. The **event research agent** retrieves evidence and proposes event–asset mappings for review. The **analysis coordinator** submits validated calculation requests. The **explanation agent** presents saved results and answers questions with source references. Section 2.2 specifies the proposed CrewAI, retrieval and tool implementation.

For “How could the next rate decision affect my SPY and MSFT exposure?”, the assistant resolves the assets, horizon and analysis time. It retrieves an eligible event and proposes an event–asset mapping explanation, but numerical responses must come from historical estimates or reviewed stress assumptions.

Guided research journey and interface

A user selects an asset or hypothetical portfolio and chooses how far ahead to examine risk. EventLens presents relevant events, compares market probabilities with experimental wallet-informed estimates, and lets the user inspect wallet activity before exploring scenarios. The assistant explains the evidence, and saved reports preserve the analysis.

For example, a user researching SPY over five trading days could inspect an upcoming rate decision, identify which tracked wallets are increasing YES or NO exposure, and compare the resulting risk scenarios. An incompatible announcement horizon or unsupported asset-response estimate will be explained before the affected calculation is withheld.

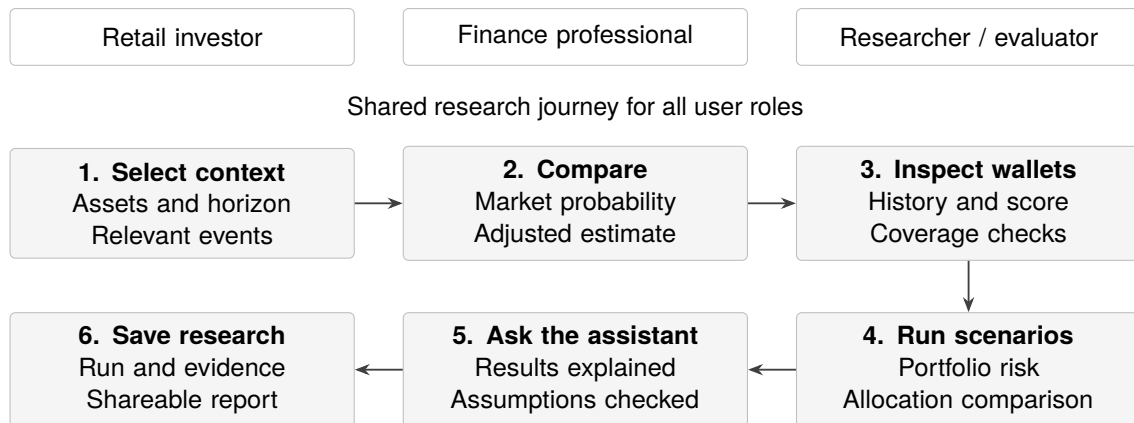


Figure 3: A representative research journey. Users can revisit any step and ask the assistant throughout.

Smart-wallet tracking and evidence inspection

Users can browse wallets in covered events, filter by history and activity, and follow public addresses without connecting a wallet. Profiles show earlier resolved-event decisions, entry prices, subsequent price movements, reconstructed positions and an activity timeline. A provisional “smart” label denotes historically informative behavior and is supported by a dated score, sample size and coverage assessment.

Concept screens

Figures 4 and 5 illustrate the intended layout. They are conceptual interface images. Values are illustrative and do not represent research results.

The **Overview** screen combines the **Watchlist** with the selected event's **Market probability**, **Wallet-informed probability**, **Evidence** status and **Probability history**. The **Research assistant** answers contextual questions and links to the **Probability run** and **Wallet evidence**. The **Events** and **Saved research** navigation items lead to detailed event review and preserved analyses.

The **Scenarios** screen is headed **Portfolio scenario analysis**. Users enter a **Paper portfolio**, select **Run scenario**, and review **95% VaR**, **95% expected shortfall** and **Event probability**. They can then choose **Compare allocation**, **Inspect evidence** or **Save report**. Inputs and results remain hypothetical and do not place orders.

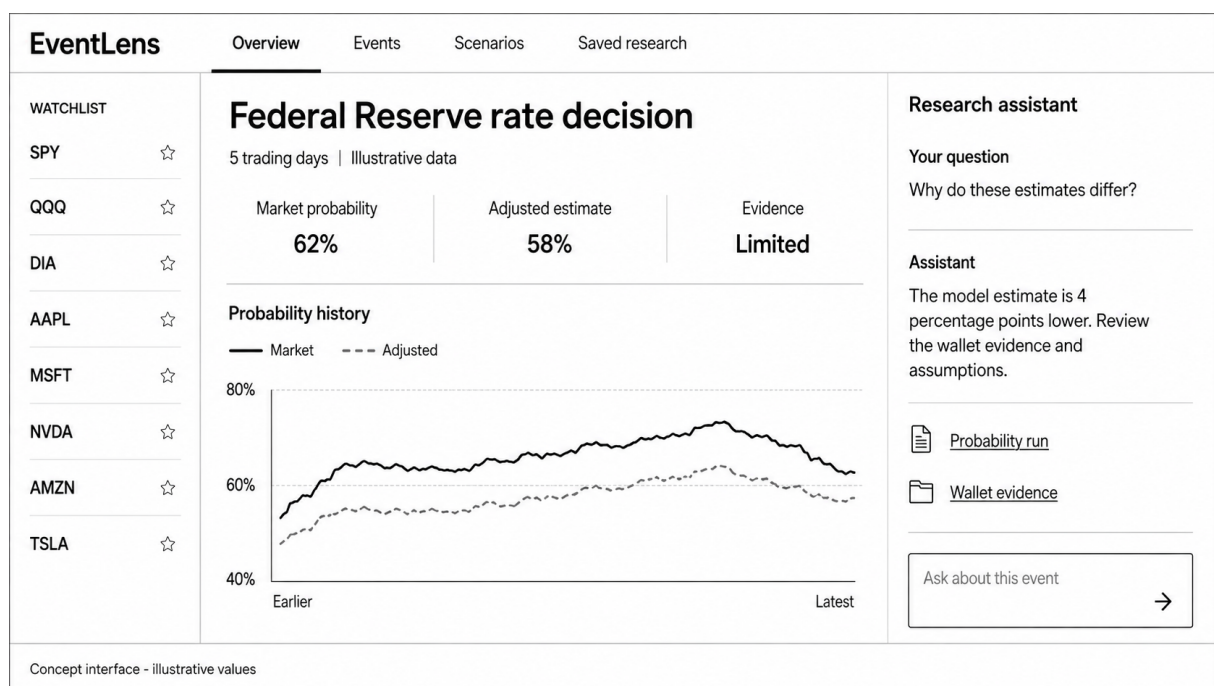


Figure 4: Overview concept with the **Watchlist**, probability comparison, **Evidence** and **Research assistant**. All displayed values are illustrative.

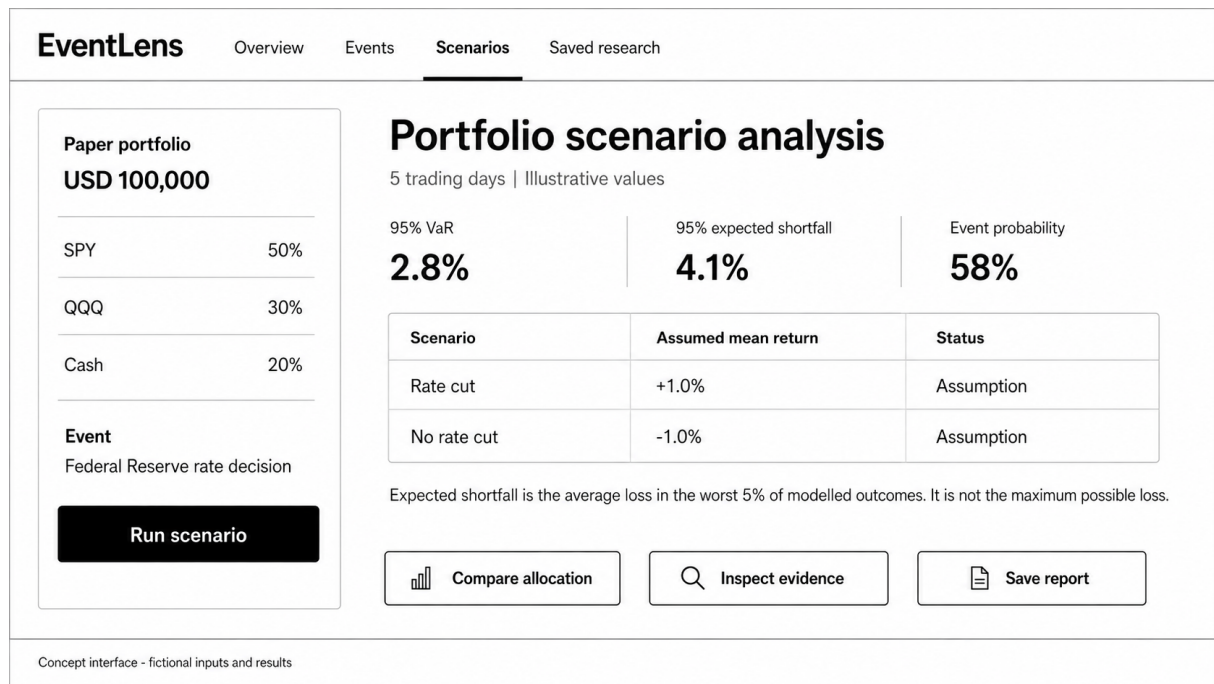


Figure 5: Scenarios concept with a **Paper portfolio**, risk measures, assumptions and comparison actions. The user explores hypothetical results without placing orders.

2.2 Implementation

The implementation below is a proposed starting configuration. Named libraries, schemas and model settings will be explored during the feasibility stage.

Application structure

Next.js and React with TypeScript will use TanStack Query for data fetching and job polling [21]. FastAPI and Pydantic will validate the JSON contracts [20]. SQLAlchemy and Alembic will manage database access and migrations. A Python worker will collect data, fit models and run scenarios.

API and frontend integration

Table 4: Proposed endpoints and compact Pydantic contracts.

Endpoint	Input schema	Response schema
GET /events	asset, as_of, cursor	events[], next_cursor
GET /wallets/{address}	event_id, as_of	score, coverage, activity[]
POST /analyses	AnalysisRequest (below)	202: job_id, status
GET /jobs/{id}	job UUID	status, result?, error?
GET /runs/{id}	run UUID	metrics, units, as_of, evidence[]
POST /assistant/query	session_id, question, run_id?	202: job_id, status

AnalysisRequest: event_id: UUID, as_of: UTC timestamp, horizon: 1|5|20, holdings: [{asset: string, weight: float}]. Weights must be non-negative and sum to one. Invalid inputs return 422 with field errors. AssistantAnswer contains text, run IDs and citations. Cache keys include input and data/model versions. Stale or incomplete evidence is flagged.

Data ingestion and storage

HTTPX collectors will fetch incremental pages. Polars/PyArrow will preserve normalized Parquet evidence. Checkpoints, retries and unique source keys will support safe replay. Fills, transfers, splits, merges and redemptions will be reconciled, with sampled Polygon receipt checks.

Table 5: Core physical schema. PK = primary key, FK = foreign key. Internal IDs use UUIDs.

Table	Key fields and constraints
market_snapshot	id PK, market_id FK, observed_at/collected_at timestamps, price numeric(18,8), evidence_uri text, source_key text unique.
wallet_score	id PK, address text, event_id FK, cut_off timestamp, score float, history_count integer, version text.
analysis_run	id PK, event_id FK, request JSONB, snapshot_ids JSONB, model_version FK, seed integer, results JSONB.
model_version	id text PK, estimator text, params JSONB, train_cut_off timestamp, artifact_uri text.
job	id PK, run_id FK nullable, kind/status text, attempts integer, heartbeat timestamp, error JSONB.
evidence_chunk	id PK, event_id FK, source_uri text, body text, available_at timestamp, embedding vector(384), embedding_version text.
assistant_session	id PK, owner_id FK, summary text, context JSONB, updated_at timestamp.

Market/time and wallet/event/cut-off indexes support dated lookups. Numeric results stay in SQL/Parquet. Embeddings support document retrieval only.

Analytical pipeline

Table 6: Example feature spaces. Each row is constructed at an information cut-off t .

Task / one row	Candidate features X_t	Target / output
Volatility / asset-date	Past 20-day mean r^2 , HAR daily/weekly/monthly components, raw event probability, adjusted minus raw probability ($p_{\text{adj}} - p_{\text{raw}}$), probability jump (Δp_{adj}), VIX/MOVE baselines	Future $RV_{t,H}$ for $H = 1, 5, 20$
Probability / event-date	Raw probability, spread, anonymous directional flow, eligible-wallet weighted flow, fresh-wallet anomaly flow	Resolved YES/NO label, then p_{adj}
Clustering / wallet-date	Turnover/capital, net YES–NO exposure, event concentration, typical holding duration	Behavior group, with no outcome label

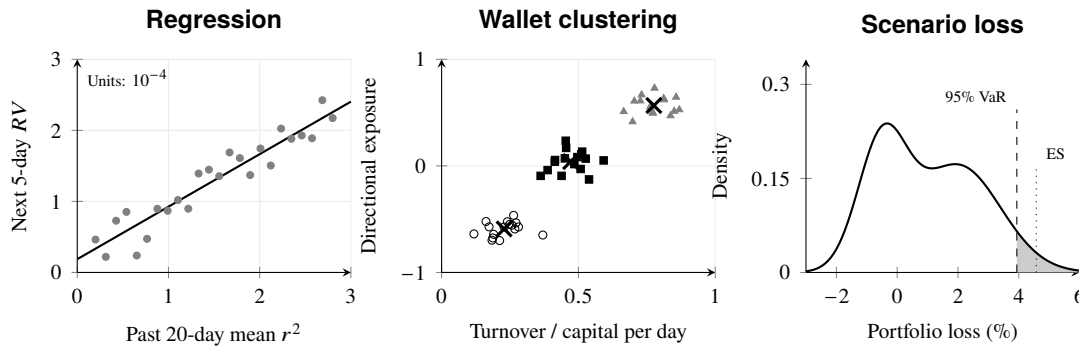


Figure 6: Synthetic examples, not research results. A fitted line, wallet groups with centroid crosses, and a loss mixture with its worst 5% shaded. ES is the mean loss in that tail.

Table 7: Candidate implementations in scikit-learn, statsmodels/arch, and XGBoost. The team will explore all techniques and retain models supported by validation.

Technique	Estimator / role	Initial exploration
Linear regression	LinearRegression / variance	Recent-return baseline and event-augmented fit.
HAR-RV & GARCH-X	OLS / GARCH(1,1) with Volatility Process	Multi-frequency HAR components and conditional variance modeling augmented with exogenous prediction-market signals (p_{adj} , Δp_{adj}).
Logistic regression	LogisticRegression / probability	Regularization strength and wallet-feature contribution.
Random Forest & XGBoost	RandomForestRegressor / XGBRegressor	Non-linear feature interactions across probability spreads, liquidity controls, and macro regime indicators using QLIKE/MSE loss objectives.
SVM	SVC / outcome, SVR / variance	Linear or RBF kernel, C , gamma. Calibrate classifier scores chronologically.
MLP	MLPClassifier / MLPRegressor	One small hidden layer, e.g. 8 or 16 units, L2 penalty.
Clustering	StandardScaler + KMeans	Explore $k = 2-5$, separation and stability across dated samples.

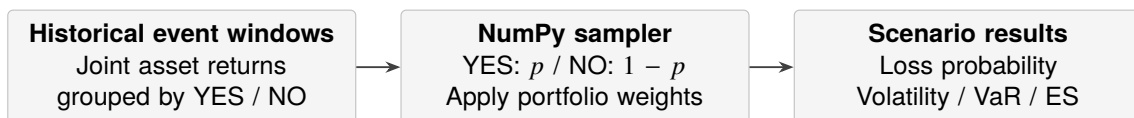


Figure 7: Scenario execution preserves joint asset returns. The seed and evidence manifest are saved.

Experiment controls. The pipeline will fit scaling and imputation procedures on the training folds only. Chronological, event-grouped validation will purge overlapping targets, including for GARCH-X, Random Forest, XGBoost, and MLP tuning. Scores, clusters and probability fits use past data only. Model evaluation for volatility forecasts will report standard robust loss functions (e.g., QLIKE, MSE) alongside directional accuracy. The regression baseline stays fixed. Clusters do not establish skill or ownership. Incomplete positions exclude affected features. Insufficient scenario evidence triggers an unavailable result or a labeled, reviewed stress assumption.

Research assistant integration

CrewAI Flows is the proposed orchestrator. A typed Pydantic state will hold session_id, event_id, assets, horizon, as_of, evidence IDs and run IDs. Three agent roles will work inside an explicit retrieve–calculate–explain flow. The workflow graph controls execution. Event–asset mappings will initially use relational joins, with knowledge-graph retrieval reserved for later exploration.

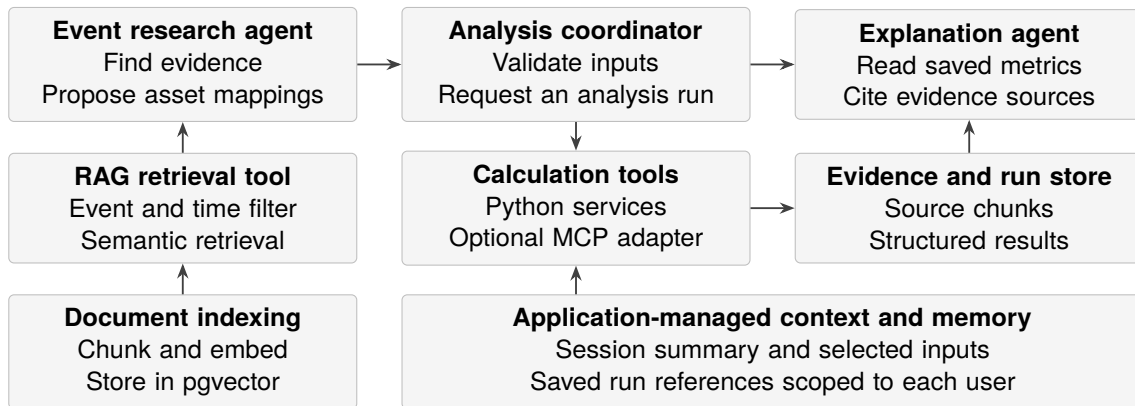


Figure 8: Proposed assistant workflow. Retrieved prose provides context. Typed calculation outputs provide numerical claims. New event–asset mappings require review before use.

Table 8: Concrete retrieval, tool and memory design choices.

Mechanism	Proposed implementation
RAG + embeddings	Archive event rules, official releases and reviewed mappings. Split the documents into chunks of approximately 180 model tokens, with an overlap of 30 tokens. The Sentence Transformers model all-MiniLM-L6-v2 produces 384-dimensional embeddings. Store source, date and embedding version with every chunk.
Retrieval	pgvector cosine search plus PostgreSQL keyword search. Merge ranked matches and return an initial set of five results. Filter by event, user access and information cut-off before ranking. Exact prices and positions come from SQL tools.
Function / tool use	CrewAI BaseTool with Pydantic arguments: search_evidence(query, event_id, as_of), get_wallet_evidence(address, event_id, as_of), run_analysis(AnalysisRequest), get_run(run_id). Tools return evidence IDs, job IDs or typed results.
MCP option	A CrewAI MCP adapter may expose the same approved functions to other clients. MCP is a tool interface, not a memory store. Direct Python tools are sufficient for the initial single deployment.
Context + memory	Load selected inputs, the latest six messages, a session summary, retrieved chunks and referenced results into a bounded prompt. Store owner-scoped summaries/preferences in PostgreSQL. Reload numerical facts from run IDs. Summaries never become research evidence.

Deployment and reproducibility

Docker Compose will run frontend, API, worker and PostgreSQL/pgvector services. A PostgreSQL job queue with retries and heartbeat expiry will recover interrupted work. Record prompts, tool traces, embedding/model versions and source manifests. Keep authenticated access, secrets, persistent volumes and backups. The assistant will stop after bounded tool calls, report missing evidence and preserve dashboard access when the hosted model fails.

2.3 Testing

Testing checks that the implementation is *correct*, that the code and pipeline do what they are meant to do – independently of whether the underlying research hypothesis turns out to be true.

Data and pipeline tests. Ingestion will be checked for schema validity, duplicate-free replay after an interrupted collection run, and reconciliation of a sample of collected records against source APIs and blockchain data. A point-in-time check will reject any feature or wallet history that uses information published after its forecast cut-off, since look-ahead leakage is the failure most likely to silently invalidate every downstream result without producing an obvious error. Corporate-action handling (splits, dividends) and timestamp normalization across time zones and daylight-saving changes will also be covered, since these are easy to get subtly wrong and hard to notice later.

Calculation tests. The probability model, wallet-scoring routine and the scenario/risk engine will be unit-tested against reference cases with known outputs: event probabilities of exactly zero and one, a missing asset, a one-sided or illiquid order book, and a singular or near-singular covariance matrix. API-level tests will enforce structural invariants – portfolio weights non-negative and summing to one, horizons restricted to supported values, VaR and expected shortfall computed consistently with their definitions – so that a broken invariant fails loudly rather than producing a plausible but incorrect result.

Workflow and agent tests. End-to-end tests will exercise the full path from event selection to a saved, shareable report, including stale or unavailable upstream data. Agent-specific tests will cover malformed or out-of-scope tool requests and adversarial instructions embedded in retrieved market or news text, since retrieved content is treated as untrusted data rather than instructions. The standing requirement is that every number shown to the user, including any number referenced by the AI assistant, traces back to a saved, versioned result rather than to generated text. This will itself be checked by a small set of tests that compare assistant output against the underlying stored result.

Indicative metrics. Pass/fail against the reference cases and invariants above, automated-test coverage of the ingestion and calculation modules, zero duplicate records after a replayed or interrupted collection run, and successful, byte-identical reproduction of a prior run's results from its stored data manifest, configuration and random seed.

2.4 Evaluation

Evaluation asks whether the finished system meets the objectives in Section 1.2.

Table 9: How each objective will be evaluated (indicative and subject to refinement after the feasibility audit).

Objective	Question	Indicative metric
Wallet identification	Does the smart-wallet score capture information beyond chance?	Backtested accuracy/Brier score vs. a label-shuffled baseline, with sensitivity to the eligibility threshold
Wallet-informed probability	Does wallet-weighting improve on the raw market price?	Brier score and log loss across the baseline ladder (raw price → +liquidity → +anonymous flow → +wallet signal)
Volatility forecasting	Do event features improve the same regression model?	MSE and QLIKE for the recent-return baseline and its augmented version on identical forecast dates
Related-event analysis	Do reviewed related events add information beyond the target event?	MSE and QLIKE for target-only versus target-plus-related-event features on identical forecast dates
Agent contribution	Does agent-assisted event selection improve over fixed rules?	Mapping precision and downstream forecast loss for rule-based, agent-assisted and hybrid configurations
Dashboard and assistant	Is the system usable, and does it stay evidence-linked?	Formative user study: task completion, numerical agreement between assistant and dashboard, correct abstention on insufficient data

Experimental design

Comparisons will use a chronological train/validation/test split, with each event’s contracts and snapshots kept together and overlapping return windows purged at split boundaries, so that no test-period information leaks backward. The final test interval will remain untouched until the models and thresholds are fixed. Wallet scores, calibration and scenario inputs will use only information available as of each historical forecast time, matching the point-in-time discipline enforced in testing.

Predictive stability will also be reported across rolling and calendar subperiods. This will reveal whether an apparent improvement is persistent or concentrated in an earlier period. A signal that performs well initially but disappears in later unseen data will be reported as unstable rather than presented as general predictive value.

Given the likely small number of independent announcement dates, results will be reported with confidence intervals rather than as single numbers, and a broader macro sample would require a documented scope revision rather than an informal expansion mid-project. Multiple wallet-level significance tests will use a false-discovery correction rather than being read individually. A null result – no measurable improvement from wallet or event signals – is a valid and reportable outcome. Profitability is explicitly not a success criterion.

Non-functional evaluation

Beyond whether the numbers are right, the dashboard needs to be evaluated on qualities that determine whether it is actually usable for research in practice.

Data freshness. The interface implies near-real-time market and wallet state, so freshness itself needs to be measured rather than assumed: the lag between a Polymarket price update or a new on-chain fill and its appearance on the Overview and Event detail screens, and the lag between an FOMC announcement and the corresponding event being marked resolved. Data is never truly instantaneous once indexing and reconciliation are involved, so the target is a bounded, disclosed lag rather than zero latency – the evidence

panel already surfaces an as-of time for this reason, and that timestamp's accuracy is itself something to check, not just display.

Responsiveness. Interactive screens (watchlist, event detail, scenario studio) should remain usable while heavier jobs run in the background. Response time will be measured separately for cached/precomputed views versus a fresh scenario run or wallet backfill, since the latter are expected to be slower and are already designed to return a job identifier rather than block the interface.

Reliability. Uptime and error rate of the deployed prototype will be logged informally over the evaluation period (e.g., during the formative study and demonstration), along with the frequency of upstream provider failures (rate limits, stale feeds) and whether the system degrades to a clear "data unavailable" state rather than showing an unlabeled stale number.

Cost. Hosted-model and data-provider cost per research query and per full wallet backfill will be tracked, since this bounds how often scores can realistically be refreshed and how many events the assistant can be exercised against during the evaluation itself.

These are secondary to the correctness and research findings above, and will be reported descriptively (typical and worst-case figures) rather than against hard service-level targets, which would not be meaningful for a research prototype at this scale.

Success requires reproducible data and experiment configuration, a controlled baseline comparison, a tested risk engine, and a usable, evidence-linked interface. Positive, negative and null findings will all be reported. Exact sample sizes, thresholds and the final participant count will be confirmed once the data-feasibility audit (Section 2.1) fixes what history is actually available.

3 Project Planning

3.1 Division of Work

Franklin, Anthony, Owen and Sunny will work across four tracks, each with a lead and a reviewer. Named ownership remains to be agreed. All members will contribute to literature review, reporting and presentations.

Table 10: Proposed work allocation, with named ownership to be recorded by the team.

Track	Responsibility	Deliverable
Data and blockchain	Source adapters, collection, reconciliation and quality reporting.	Versioned dataset and coverage report.
Probability research	Wallet scoring, calibration, baseline comparisons and leakage controls.	Reproducible probability experiments.
Scenarios and risk	Asset-response estimates, volatility, tail risk and reference tests.	Deterministic risk engine and application study.
Product and AI	Backend integration, frontend, agent tools and deployment.	Integrated dashboard and user evaluation.

Git will hold code, experiment configuration and separate LaTeX section files for collaborative editing. Generated files and secrets stay out of version control. Weekly reviews and an issue tracker will record owners, dependencies, meeting decisions and experiment changes.

3.2 GANTT Chart

The schedule is relative to project commencement and will be aligned with course dates. Data feasibility comes first. Interface work starts with sample data to identify integration problems early.

Table 11: Planning baseline. Shaded cells indicate the main work period.

Workstream	1–4	5–8	9–12	13–16	17–20	21–24
Scope, literature and feasibility						
Collection and reconciliation						
Wallet and probability models						
Event mapping and risk engine						
Interface and AI integration						
Evaluation and user study						
Report and reproducibility						

Milestones. Week 2: initial access and scope decision. Week 7: a versioned dataset and reconciliation report. Week 12: first raw-price, anonymous-flow and wallet-flow comparison. Week 18: working scenarios and risk calculations. Week 22: integrated evaluation and user tasks. Week 24: final report, demonstration and reproducibility package. Failed feasibility gates will narrow the model or coverage before optional work begins.

4 Hardware and Software Requirements

4.1 Hardware Requirements

The initial models and API-hosted language model do not require a local GPU. Development can be carried out on standard personal computers, with a modest shared server for scheduled collection and the demonstration.

Table 12: Proposed hardware provision.

Resource	Minimum provision	Preferred provision
Developer computer	Four-core CPU and 8 GB RAM	16 GB RAM for concurrent local services.
Storage	30 GB free disk space	Approximately 100 GB SSD space for bounded historical data.
Shared server	Local demonstration if necessary	2–4 vCPU and 8–16 GB RAM, subject to measured workload.
Network	Stable internet connection	Reliable access for collectors, hosted models and remote collaboration.

A self-hosted Polygon node is outside the initial scope. Hosted RPC and targeted indexed data will be used for collection and verification. Database and raw-data storage must persist across application updates and be backed up regularly.

4.2 Software Requirements

Table 13: Required software and external services.

Category	Software or service	Purpose
Runtime	Python and Node.js	Research services, collectors and web application.
Application	FastAPI, Pydantic, Next.js/React, TypeScript and TanStack Query	Validated backend APIs and interactive interface.
Data and modeling	PostgreSQL/pgvector, SQLAlchemy, Parquet, Polars, NumPy and scikit-learn	Storage, processing and initial statistical models.
Assistant candidates	CrewAI, Sentence Transformers and optional MCP adapter	Agent workflow, evidence retrieval and typed tool access.
External services	Polymarket APIs, Goldsky, Polygon RPC, equity data and hosted LLM	Event evidence, asset histories and research assistance.
Quality and deployment	Pytest, browser tests, Docker and Docker Compose	Automated verification and reproducible deployment.
Collaboration	Git, issue tracker and LaTeX	Source control, task allocation and shared documentation.

External costs may include adjusted equity histories, indexed blockchain access, archival RPC, a hosted language model and server hosting. Subject to final confirmation of the currency and available funding, the project will plan for an indicative external-service budget of 1,000–2,000. Spending will prioritize historical market data, indexed blockchain access and hosted-model usage. Costs will be reviewed during the feasibility stage, with free-tier or locally hosted alternatives used where practical. A paid

historical options dataset is not required for the core project. Software dependencies, model versions and configuration will be recorded to support reproducibility.

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6 Appendix A: Meeting Minutes

6.1 Minutes of the 1st Project Meeting

Date: 8 August 2026 (Saturday)

Time: 21:00

Place: Zoom

Present: All members

Absent: None

Recorder: Sze Wang Cheong

1. Approval of minutes

There were no minutes to approve, as this was the first formal group meeting.

2. Report on progress

2.1 All members had reviewed the FYP proposal template and familiarized themselves with the submission requirements.

2.2 The group discussed the overall timeline and key milestones leading up to the proposal deadline.

3. Discussion items

3.1 The group agreed on a structured timeline before the proposal submission: idea-sharing on 16 August, buffer sessions on 22 and 29 August, internal deadline (draft to communication tutor) on 3 September, review of tutor comments on 6 September, soft submission deadline on 10 September and hard submission deadline on 11 September.

3.2 The objective of the next meeting (16 August) was defined: each member would prepare and share project ideas, with the group deciding on a final topic by the end of that meeting.

4. Goals for the coming month

4.1 All members to prepare project ideas for the next meeting.

4.2 The group to keep to the agreed timeline so the internal draft is ready for tutor review by 3 September.

5. Meeting adjournment and next meeting

The meeting was adjourned at 21:12. The next meeting was scheduled for 21:00 on 16 August 2026, via Zoom.

6.2 Minutes of the 2nd Project Meeting

Date: 16 August 2026 (Sunday)

Time: 21:00

Place: Zoom

Present: Anthony, Sunny, Owen, Franklin

Absent: None

Recorder: Team representative

1. Approval of minutes

The minutes of the 1st meeting (8 August 2026) were reviewed and approved.

2. Report on progress

2.1 All members had researched candidate FinTech, Web3 and quantitative trading topics as scheduled.

2.2 All members prepared proposal ideas to present at this meeting.

3. Discussion items

3.1 Each member presented candidate directions: Anthony proposed several crypto/quant concepts, including cross-exchange arbitrage, an on-chain whale-tracking agent, a crypto risk-management framework and a market-anomaly-detection agent. Sunny proposed an end-to-end quant platform with an NLP module to translate strategy descriptions into backtestable code. Owen supported Sunny's direction, citing WorldQuant Brain as an industry reference. Franklin proposed an on-chain smart-wallet tracking system on Polymarket (Polygon) using a scoring model to detect informed trading.

3.2 The team agreed the project should be product-centric rather than strategy-centric, to keep a strong engineering and system-design scope.

3.3 The team noted the NLP-backtesting idea risked being too general and hard to backtest reliably for crypto/Web3 dynamics.

3.4 The team reviewed the feasibility of Franklin's Polymarket proposal, noting existing open-source tools already demonstrate reliable extraction of Polymarket on-chain data.

3.5 The team reached consensus on a unified direction: a Web3 intelligence system that ingests on-chain Polymarket data, identifies smart-wallet activity on relevant prediction markets, and translates those signals into insights for financial and equity markets (e.g., macro risk indicators).

4. Goals for the coming week

4.1 All members to research the methodology and architecture needed for the Polymarket intelligence platform (data sources, APIs, on-chain indexing and signal mapping).

4.2 All members to prepare detailed technical inputs for the draft proposal ahead of the internal deadline on 3 September.

5. Meeting adjournment and next meeting

The meeting was adjourned at 22:56. The next meeting was scheduled for 21:00 on 22 August 2026, via Zoom, to discuss system methodology and data-pipeline architecture.

6.3 Minutes of the 3rd Project Meeting

Date: 22 August 2026 (Saturday)

Time: 21:00

Place: Zoom

Present: All members

Absent: None

Recorder: Team representative

1. Approval of minutes

The minutes of the 2nd meeting (16 August 2026) were reviewed and approved.

2. Report on progress

2.1 Members had used AI assistance to generate an initial full draft of the proposal (Overview, Objectives and Methodology) based on the team's prior discussions and research.

2.2 The draft was found to be substantially over length relative to the expected proposal size.

3. Discussion items

3.1 The team reviewed the proposed user workflow: a user selects a supported market category (e.g., a Federal Reserve rate-cut event), the system applies smart-wallet identification to derive a wallet-informed probability, and an optional simple backtest of a volatility relative-value strategy is run on the result.

3.2 The team discussed the intended MVP dashboard: user-selected event categories, intermediate outputs (smart wallets, wallet-informed probability) and final outputs (volatility and risk metrics such as value at risk (VaR)).

3.3 The team clarified the role of the LLM agent: summarizing smart-wallet evidence for the user, mapping a user's watchlist or portfolio to relevant Polymarket events, and offering context-appropriate suggestions once the user has entered their holdings.

3.4 The team confirmed all on-chain and market data would be sourced from Polymarket's API, including its central limit order book.

3.5 Referring to the FYP grading rubric (approximately 60% objectives/methodology/background, 30% organization/clarity/presentation, 10% planning and timeline), the team agreed the draft needed significant shortening, targeting roughly 10–15 pages, with the Design and Application subsection cut most aggressively.

3.6 The team discussed collaborative tooling for the LaTeX draft (Overleaf's per-project collaborator limits versus a Git-based workflow) and agreed to continue evaluating options.

3.7 The team agreed to request a consultation with the supervisor (Prof. Shuai Wang) ahead of submission, proposing Monday afternoon or Tuesday morning as candidate slots.

4. Goals for the coming week

4.1 Anthony to shorten the Introduction (overview and objectives).

4.2 Owen and Franklin to jointly shorten the Design and Application sections.

4.3 The remaining member to shorten the Testing and Evaluation section.

4.4 Franklin to email the supervisor to arrange a consultation slot before the internal deadline.

5. Meeting adjournment and next meeting

The meeting was adjourned. The next meeting was scheduled for 21:00 on 29 August 2026, via Zoom.

6.4 Minutes of the 4th Project Meeting

Date: 29 August 2026 (Saturday)

Time: 21:00

Place: Zoom

Present: Anthony, Sunny, Owen, Franklin (Sunny joined shortly after the meeting began)

Absent: None

Recorder: Team representative

1. Approval of minutes

The minutes of the 3rd meeting (22 August 2026) were reviewed and approved.

2. Report on progress

2.1 Members had continued drafting their assigned sections following the shortening exercise agreed at the previous meeting.

3. Discussion items

3.1 The team revisited smart-wallet identification in more depth: given the limited history available on Polymarket (approximately three years) and the absence of a labeled dataset, the team agreed to first define rule-based labels (e.g., a historical win-rate threshold) and then train a model on these labels so that new, previously unseen wallets can also be scored.

3.2 The team discussed mapping the wallet-informed probability to asset volatility, identifying GARCH-family models as a candidate approach for incorporating the wallet-informed probability as an input feature, while noting this mapping step has fewer observations per event than the wallet-scoring step and needs a simpler, more robust design.

3.3 The team agreed to narrow the initial asset universe to a small set of index-tracking ETFs, which carry less idiosyncratic risk and are easier to forecast, before considering individual equities in a later phase.

3.4 The team clarified that the machine-learning components (wallet scoring, probability-to-volatility mapping) and the LLM research agent (event-asset mapping and user-facing explanation) are complementary rather than overlapping, so the design satisfies the project's AI requirement without redundancy.

3.5 The team discussed two possible uses for the volatility output – portfolio risk management versus an actively traded volatility strategy – and agreed to describe both directions in the proposal while prioritizing the risk-management framing, since the primary target user is a retail investor rather than a derivatives trader.

3.6 The team agreed to source on-chain settlement data via Polygon RPC, in addition to Polymarket's own API for event metadata, to keep the project's Web3 focus. Initial event coverage will prioritize finance/macroeconomics and politics/geopolitics categories over sports or other categories.

3.7 The team reviewed the FYP table of contents and grading rubric and split the remaining proposal-writing workload by section among the four members, agreeing to draft independent sections individually first and reconvene to jointly resolve the more interdependent Design and Methodology content.

3.8 The team noted an open question requiring further work: how to weight and aggregate multiple related Polymarket bets into a single wallet-informed volatility estimate for a given asset.

4. Goals for the coming week

4.1 All members to continue drafting their assigned proposal sections.

4.2 Members to individually explore a small proof-of-concept calculation for the probability-to-volatility aggregation step, to compare approaches at the next meeting.

5. Meeting adjournment and next meeting

The meeting was adjourned. The next meeting was scheduled ahead of the internal deadline of 3 September 2026, date and time to be confirmed.

6.5 Minutes of the Supervisor Consultation

Date: 10 September 2026 (Thursday)

Time: 10:00

Place: Online meeting

Present: Prof. Shuai Wang and the project team

Recorder: Team representative

1. Review of the proposal

1.1 The supervisor was satisfied with the team's progress and considered the proposal report suitable for submission after final refinement.

1.2 The supervisor did not confirm a preference for passive voice. The team therefore agreed to prioritize clear and consistent academic writing.

2. Suggestions

2.1 The supervisor suggested investigating relationships among Polymarket events. Events that are not directly selected for analysis may still contain related information through shared topics, entities or outcomes.

2.2 The supervisor suggested an ablation or comparison study between fixed rule-based signals and analysis assisted by an LLM agent, with a hybrid approach as an additional configuration if capacity permits.

2.3 The team noted that predictive relationships may decay over time and agreed that any proposed signal should be assessed across chronological and rolling evaluation periods.

2.4 The supervisor clarified that the cross-event and agent-ablation studies were suggestions rather than required additions and could be prioritized according to available time and capacity.

3. Resources

3.1 The team asked about funding for external APIs and related services. The supervisor indicated that a total budget in the range of 1,000–2,000 would be reasonable, subject to confirmation, with a possible limited top-up if justified later.

4. Actions

4.1 The team will refine the proposal to record related-event analysis and agent ablation as bounded extensions rather than unconditional MVP commitments.

4.2 The team will confirm the approval process before incurring external-service costs.

4.3 The team will complete language and consistency checks before submission.